

Task 4:- PWM Test

4_pwm.ino

```
1  // ESP32 PWM on Pin 23 (Core v3.x compatible)
2
3  const int pwmPin = 23;      // PWM output pin
4  const int pwmChannel = 0;   // Channel 0-7
5  const int pwmFreq = 5000;  // Frequency in Hz
6  const int pwmResolution = 8; // Resolution bits (0-255 for 8-bit)
7
8  void setup() {
9      Serial.begin(115200);
10
11      // Attach PWM channel to pin
12      ledcAttach(pwmPin, pwmFreq, pwmResolution); // <-- new API in ESP32 core v3.x
13      Serial.println("PWM example started!");
14  }
15
16  void loop() {
17      // Sweep duty cycle from 0 to 255
18      for (int dutyCycle = 0; dutyCycle <= 255; dutyCycle++) {
19          ledcWrite(pwmPin, dutyCycle); // write duty to pin
20          Serial.print("PWM Duty Cycle: ");
21          Serial.println(dutyCycle);
22          delay(50);
23      }
24      delay(1000);
25      // Sweep back from 255 to 0
26      for (int dutyCycle = 255; dutyCycle >= 0; dutyCycle--) {
27          ledcWrite(pwmPin, dutyCycle);
28          Serial.print("PWM Duty Cycle: ");
29          Serial.println(dutyCycle);
30          delay(50);
31      }
32  }
33
```